

Michelle Herrera-Cuen

Computer Engineering Student | Embedded Systems | Robotics & Control | Firmware Development

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Technical Skills: C/C++, Python, Verilog, Microcontrollers, Robotics Simulation, Git, Linux, Altium Designer

Education

Bachelor of Science in Computer Engineering

California State University, Long Beach | Expected 2027

Experience

Honors Thesis: Configurable Robotics Firmware Platform

Undergraduate Research Assistant Long Beach, CA | Jan 2025 - Present

Designed a lightweight robotics ecosystem inspired by PX4/Pixhawk enabling no-code/low-code deployment for ground, marine, and articulated robots. Built a desktop GUI to define robot topology, sensors, and kinematic constraints; auto-generated and flashed firmware to ESP32, Teensy, Arduino, and STM32 MCUs. Architected modular abstraction layers for sensor integration, communication (I2C, SPI, UART), telemetry, PID tuning, and calibration

Bio-Inspired Robotic Fish Research | CSULB PACK Lab

Undergraduate Research Assistant Long Beach, CA | Jan 2025 - Present

Modeled fish-like underwater propulsion using Python and embedded C, analyzing tail curvature, frequency, stiffness, and hydrodynamic forces. Developed numerical simulations and control algorithms for energy-efficient swimming and autonomous stability.

Cybersecurity Intern

Blizzard Entertainment — Los Angeles, CA | Summer 2024

Supported cybersecurity workflows and vulnerability analysis in a large-scale production environment.

Hardware Engineering Intern

DMC Engineering — Los Angeles, CA | Summer 2025

Supported hardware and controls development for industrial automation systems, including system integration and testing. Collaborated with controls and software engineers to troubleshoot hardware, sensors, and electrical interfaces

Additional Experience: Team Lead, Jack in the Box (2022–Present)

Projects

Embedded Space Invaders Game

Developed an embedded game in C on a TM4C LaunchPad. Integrated Nokia 5110 LCD graphics, ADC-based potentiometer input, GPIO interrupts, and PWM-driven DAC audio. Implemented real-time firmware architecture with collision detection and interrupt-based timing.

Autonomous Robotics Systems

Designed object-following and wall-following robots using IR sensors, DC motors, and microcontroller-based control. Implemented UART, I2C, SPI, ADC, PWM, and interrupt-driven navigation logic.

Associations

Women in Computing (WIC)

President | Aug 2024 - Present

Lead technical workshops, hackathons, mentorship initiatives, and development of a First Place awarded Python-based educational app teaching marine science concepts during a 2025 hackathon.

Society of Automotive Engineers

Data Acquisition Team Lead | Sept 2024 - Present

Led ESP32-based data acquisition system development. Implemented LoRa communication between ESP32 and Raspberry Pi for real-time sensor data collection.

